SPURRED TORTOISE POPULATION ASSESSMENT

A Field Conservation Report on the Status of Centrochelys sulcata in Northern Ghana

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Lead Agency: WATCI / Wildlife Division (FC)

Region: Upper East, Upper West & North East

Executive Summary

The African Spurred Tortoise (***Centrochelys sulcata***), also known as the Sulcata tortoise, is the largest continental tortoise species in the world and is globally categorized as Endangered by the IUCN. Historically native to the arid Sahel and Sudanian savanna zones of sub-Saharan Africa, its southern distribution margin extends into Northern Ghana. This assessment represents a comprehensive initiative to document the distribution, habitat conditions, and population density of wild ***C. sulcata*** populations in Ghana.

Over a 12-month field campaign spanning 2025 and early 2026, researchers utilized systematic line transect sampling, intensive burrow searches, and structured community interviews across four key conservation sectors in Upper East, Upper West, and North East Regions. The findings indicate highly fragmented, low-density populations, heavily threatened by human encroachment, agricultural expansion, and seasonal bushfires. This report outlines urgent intervention steps to safeguard Northern Ghana's remaining wild tortoises.

> **Key Metric & Takeaway:** The average population density across surveyed sites was found to be extremely low, estimated at only $D = 0.04 \pm 0.01$ individuals per square kilometer. This highlights the species' critical vulnerability to local extirpation unless immediate community-led conservation frameworks are established.

1. Introduction & Conservation Status

The African Spurred Tortoise is a keystone ecological species of the semi-arid savanna. By constructing deep, extensive burrow systems—often reaching up to 15 meters in length—they provide vital thermal refugia for dozens of co-habiting species during the intense Sahelian dry season. Furthermore, as large-scale grazers, they play an essential role in seed dispersal and nutrient cycling across fragile ecosystems.

In Ghana, the species is legally protected under Schedule I of the Wildlife Conservation Regulations, which prohibits hunting or capture. However, domestic enforcement remains challenging. Due to its position at the southern dry-savanna limit, Northern Ghana represents a critical genetic corridor connecting West African populations. Climate-driven aridification has made preserving these populations a matter of regional conservation priority.

2. Survey Methodology

To obtain accurate population metrics, the assessment team adopted a hybrid methodology combining physical field surveys with local community ethnobiology. This approach was divided into three primary components:

2.1 Systematic Line Transects

At each of the selected sites, 10 linear transects of $L = 2.5$ *ext{ km}* were established. Teams of three trained observers walked these transects during peak activity hours—specifically in the early morning (06:00 – 09:00) and late afternoon (16:00 – 18:30) during the wet season (August to October), when activity is highest.

2.2 Burrow Identification & Density Estimation

Active burrows were identified by looking for characteristic crescent-shaped openings, slide marks, and fecal pellets. The distance from the transect centerline to the burrow opening was measured to calculate burrow density using the standard distance-sampling formula:

$$D_b = \Sigma n / (2 \times L \times W)$$

Where n represents the number of observed burrows, L represents the total transect length, and W represents the effective strip width.

2.3 Ethnozoological Surveys

Semi-structured interviews were conducted with 120 local stakeholders, including farmers, hunters, and community elders. These interviews provided historical context on species presence, local use (traditional medicine and consumption), and the frequency of human-tortoise encounters over the past three decades.

3. Field Assessment Results

A total of 42 active burrows and only 7 live wild individuals were physically encountered during the 12-month field campaign. This extremely low direct sighting rate confirms the nocturnal/crepuscular and highly subterranean lifestyle of the species during dry periods.

Survey Zone (Region)	Habitat Type	Active Burrows Found	Live Sightings	Est. Density (indiv/km^2)	Threat Level
Red Volta Valley (Upper East)	Riverine woodland / Shrubland	18	3	0.06	Severe (Bushfires)
Wechiau Buffer (Upper West)	Semi-deciduous savanna	12	2	0.04	High (Overgrazing)
Nakpanduri (North East)	Rocky escarpment savanna	9	1	0.03	Severe (Poaching)
Mole Buffer (Northern)	Open savanna woodland	3	1	0.01	Moderate

4. Primary Threats to Wild Populations

The rapid decline of ***C. sulcata*** populations in Northern Ghana is driven by a combination of anthropogenic pressures and environmental stressors:

- **Agricultural Expansion & Habitat Fragmentation:** The transition of natural savanna into monoculture farmland (primarily maize and cotton) has destroyed critical nesting sites and fragmented tortoise corridors.
- **Seasonal Agricultural Bushfires:** Annual fires set by farmers to clear land often trap slow-moving tortoises above ground, causing high mortality rates, particularly among juveniles.
- **Overgrazing by Cattle:** Intense cattle grazing reduces the availability of native high-fiber grasses that constitute the core of the tortoise's diet and leads to soil compaction, rendering the ground too hard for burrow excavation.
- **Illegal Wildlife and Pet Trade:** While domestic consumption has decreased, there is persistent opportunist poaching of hatchlings for sale to intermediary traders supplying the international exotic pet market.

5. Strategic Conservation Roadmap

To prevent the localized extinction of the African Spurred Tortoise in Northern Ghana, a collaborative, multi-stakeholder conservation strategy is urgently required.

5.1 Community-Based Protected Areas (CREMAs)

Establishing Community Resources Management Areas (CREMAs) in high-density zones—such as the Red Volta River Valley—will empower local communities to manage and protect tortoise habitats. Integrating tortoise conservation into local bylaws is the most cost-effective way to curb poaching and reduce destructive fires.

5.2 Controlled Seasonal Burning Schemes

Wildlife managers must work with local agricultural extensions to promote early-season, controlled cool burns in place of late-season hot fires. Early burning occurs when vegetation is partially damp, minimizing subterranean temperature spikes and giving burrows adequate protection.

5.3 Targeted Anti-Poaching and Pet Trade Controls

Enhancing patrol frequencies in known nesting zones during the hatching season (May to July) and training local rangers to recognize illegal tortoise collection points will directly disrupt trafficking pipelines.

6. Conclusion

The Spurred Tortoise stands on the brink of localized extinction in Northern Ghana. With average densities hovering near critical minimums, the wild populations cannot withstand continued habitat loss and unsustainable harvesting. However, because the species is exceptionally resilient and adapts well to community-managed buffer zones, immediate conservation actions—focused on fire management and habitat preservation—can successfully stabilize and restore this iconic savanna giant.